

IBM Employee Attrition

Understanding Why Talent Leaves

An Unsupervised Learning (K-Means) Approach to Workforce Segmentation

Prepared for: Dr. Sarah Chen, Chief Human Resources Officer, IBM Corporation

Prepared by: HR Analytics Consulting Team — Rabia, Purdue University MSBA

Deployment: Interactive K-Means Segmentation Application (Dash / Plotly)

Plotly Cloud Application: <https://dazzled-aqua-scallop-e691ac2f.plotly.app/>

Dataset: IBM_Dataset.csv — 1,470 employee records (2025)

1. Problem – Data – Insights – Deployment Framework

Problem	IBM's HR team lacks a proactive way to identify employees at elevated attrition risk across departments, job levels, demographics, and satisfaction. Goal: shift from "who left?" to "what kinds of employees are at risk?"
Data	1,470 IBM employee records (2025), 13 variables covering demographics, job characteristics, seniority, satisfaction, compensation, and Attrition. No missing values.
Insights	K-Means (k=4) on standardized features reveals four segments with attrition ranging 6.8%–21.6% vs. a 16.1% company-wide rate; early-career, low-tenure, lower-satisfaction employees are the highest-risk group.
Deployment	Interactive Dash/Plotly app deployed for HR leadership: Elbow/Silhouette for k=2–10, cluster profile KPIs and charts, and a scenario simulator assigning a hypothetical employee to the closest segment. Live at: https://dazzled-aqua-scallop-e691ac2f.plotly.app/

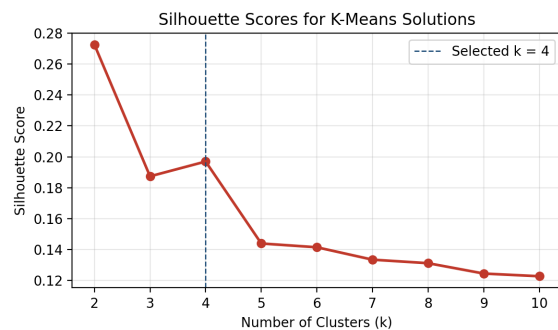
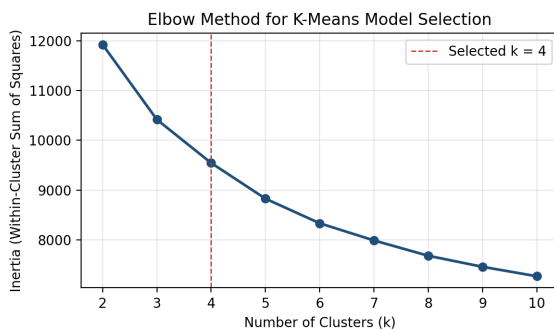
2. Data Preprocessing

All 12 explanatory variables were used for clustering. Attrition was excluded from the clustering inputs and used only afterward to calculate the observed attrition rate for each segment. Variables were treated by measurement type before K-Means:

- **Nominal** (Department, Gender) — one-hot encoded (no inherent order).
- **Ordinal** (Education, JobLevel, JobSatisfaction, WorkLifeBalance) — kept as their existing ordered integer codes, then standardized alongside continuous variables.
- **Continuous** (Age, MonthlyIncome, TotalWorkingYears, YearsAtCompany, YearsInCurrentRole, YearsWithCurrManager) — standardized (StandardScaler) so no variable dominates Euclidean distance purely by scale (e.g. income in the thousands vs. education 1–5).
- **Attrition** — excluded from clustering inputs; used only to evaluate discovered segments.

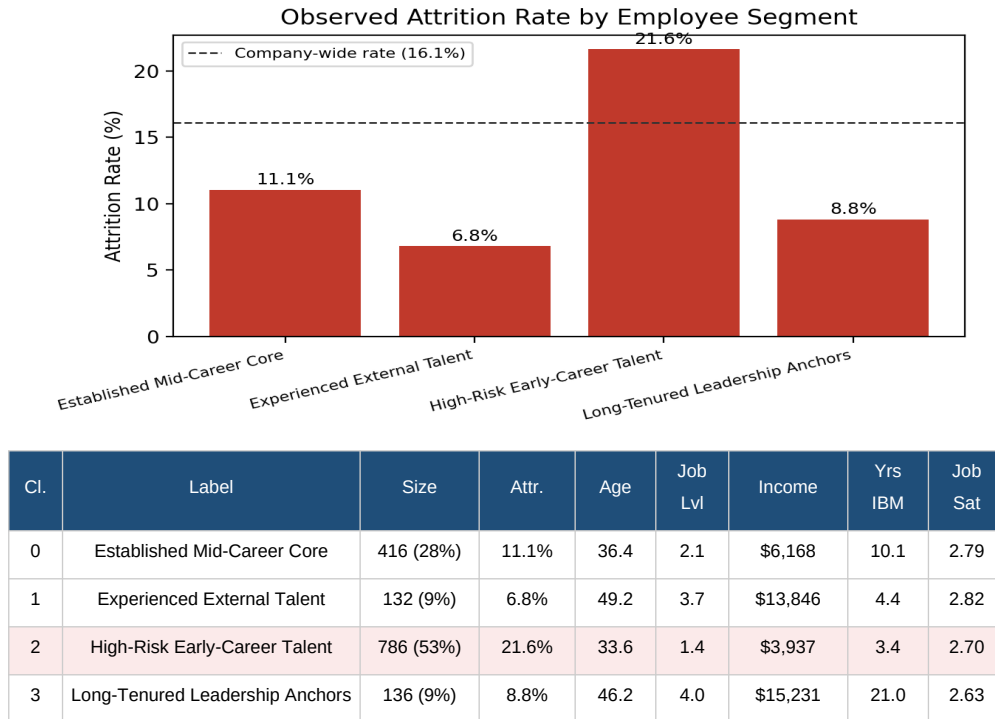
3. Clustering Quality Evaluation

Elbow Method and Silhouette Score were computed for k=2 through k=10 to justify the choice of k.



Inertia drops sharply through k=4, then flattens (elbow bend near k=4). Silhouette peaks at k=2 (0.27), but k=2 is too coarse to be actionable for HR; it rebounds locally at k=4 (0.197) versus neighbors k=3 (0.187) and k=5 (0.144). **k=4** was selected as the best balance of statistical separation and business interpretability, producing four segments that map cleanly onto distinct career stages with meaningfully different attrition rates.

4. Cluster Profile Interpretation (k=4)



Cluster 0 — Established Mid-Career Core (28%). Mid-career (avg. age 36), entry-to-junior level, moderate tenure (10.1 yrs at IBM). Attrition 11.1%, below company average — a stable, foundational group needing sustained engagement rather than urgent intervention.

Cluster 1 — Experienced External Talent (9%). Older (avg. 49), well-compensated (\$13,846/mo), high total experience (22.7 yrs) but short IBM tenure (4.4 yrs) — likely mid-career external hires. Lowest attrition of all segments (6.8%), suggesting current onboarding/comp for senior hires is working.

Cluster 2 — High-Risk Early-Career Talent (53%). Largest segment and the clear outlier: youngest (33.6), lowest job level (1.4), lowest income (\$3,937), shortest tenure (3.4 yrs), lowest satisfaction (2.70). Attrition 21.6% — over 3x Cluster 1's rate. IBM's primary retention priority given its size and risk.

Cluster 3 — Long-Tenured Leadership Anchors (9%). Most senior group: highest job level (4.0), highest income (\$15,231), longest tenure (21.0 yrs). Attrition low (8.8%). Low flight risk, but holds concentrated institutional knowledge — priority is succession planning, not volume retention.

5. Responses to Dr. Chen's Business Questions

What characterizes employees who leave vs. remain?

Departures concentrate in Cluster 2 (53% of staff, 21.6% attrition, well above the 16.1% average). Employees who stay skew toward Clusters 0, 1, and 3 — higher job level, longer tenure, higher income.

Are there demographic differences?

Gender (51–62% male) and education (avg. 2.8–3.1/5) are similar across all clusters — neither is a strong differentiator. Age is more telling: the highest-risk cluster is youngest (33.6 vs. 46–49 in low-risk senior clusters).

Are position-related factors associated with leaving?

Yes — the strongest pattern found. Job level averages 1.4 (entry) in the highest-risk cluster vs. 3.7–4.0 in the lowest-risk clusters. Department composition is relatively similar across clusters, while job level shows a much stronger association with the discovered employee segments.

Does seniority play a role?

Strongly. Years at IBM ranges from 3.4 (highest-risk cluster) to 21.0 (most senior cluster). The two most senior clusters show the lowest attrition rates (6.8% and 8.8%).

Are job satisfaction and work-life balance associated with attrition?

Directionally yes, but more modestly than job level or tenure. The highest-risk cluster has the lowest job satisfaction (2.70) and among the lowest work-life balance (2.74) scores — contributing, reinforcing factors rather than the primary driver.

6. Conclusion

K-Means segmentation reframes IBM's attrition challenge from a company-wide 16.1% statistic into a targeted problem: over half the workforce (Cluster 2) carries an attrition rate more than 3x IBM's most stable segment. Because this group is both large and high-risk, it is the highest-leverage target for retention investment. The accompanying application lets Dr. Chen's team explore these segments directly and simulate where a given employee profile falls, supporting proactive retention planning over reactive exit interviews.

Note: this analysis identifies workforce segments and their historical attrition rates. It is not an individual-level predictive model and should not be the sole basis for employment decisions about a specific person.